

SQL Syllabus

Course Duration: 40 hrs.

1	Database • What is Database or DBMS • What is RDBMS • What is Table and Entity • Types of SQL Databases • Installing Tool(MySQL, MS-SQL) and Configurations • Basic Queries
2	Java MySQL Database Access • Database Handling with MySQL or MS-SQL • Create Database Connection SQL Constrains • Primary Key • Check • Not Null • Foreign Key • Unique • Default
3	DDL Operations • What is DDL command • Creating database and Table • Dropping Database and Table • Altering the table • Using Truncate

4	DML Operations • What is DML • Inserting data into table • Updating the data in table • Deleting data or row, columns
5	DCL Operations • What is DCL • Grant • Revoke TCL Operations • What is TCL • Start transaction • Commit • Rollback DRL – Select Operations
6	SQL DataTypes • INT • DECIMAL • VARCHAR • DATE & TIME • DOUBLE or FLOAT SQL Operators • Arithmetic • Relational • Logical

7	Advanced SQL Topics SQL Clause & Keywords • Where • Group By • Having • Order By • IN • Between • Like • Distinct • Exist • Limit • Offset
8	SQL Joins • What is Joins • Inner Join • Outer Join 1. Left Join 2. Right Join 3. Full Join • Cross Join • Self-Join • Union

9	Aggregation in SQL • Count() • Sum() • Avg() • Min() • Max()
10	String Functions Math Functions Numeric Functions
11	Windows Function • ROW_Number() • RANK() • DENSE_RANK() CASE STATEMENTS
12	Subqueries in SQL • Single-row Subquery • Multiple-row Subquery • Correlated Subquery • Nested Subquery
13	SQL Programming: Stored Procedures • What is stored procedure • Creating Procedure • Using Procedure

	Functions • What is Function • Creating Function • Function Types 1. DETERMINISTIC 2. READS SQL DATA 3. NO SQL
14	Views in SQL • What is Views • Creating View tables SQL Trigger • What is Trigger • Creating before Triggers • Creating After Trigger • Using before Triggers • Using After Trigger
15	Normalization • 1NF • 2NF • 3NF • BCNF • 4NF • 5NF • DKNF
16	Exercise & Practice